POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6031T | Semester VI | Theory | 4 Credits

Microcontroller and PLC

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6031T as Microcontroller and PLC in Semester VI for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Industrial Automation and Control course and polytechnic microcontroller curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Microcontroller designs, PLC programs, industrial circuits and automation safety must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support 8051 architecture, interfacing, PLC hardware, scan cycles, and Ladder Logic. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Microcontroller and PLC studies the architecture, programming and application of embedded and industrial control systems. It develops the ability to analyze 8051 microcontroller architecture, write assembly/C programs, interface sensors and actuators, and design industrial automation solutions using Programmable Logic Controllers (PLCs) and Ladder Logic while respecting the technical and safety boundaries of electrical and industrial engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Digital electronics, microprocessor fundamentals, basic C programming and electrical machines.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the architecture, memory organization and instruction set of the 8051 microcontroller.
2. Apply assembly and C language programming to implement control logic and peripheral interfacing.
3. Explain the hardware architecture, I/O processing and scan cycle of Programmable Logic Controllers (PLCs).
4. Apply Ladder Logic programming to design and implement sequence control for industrial automation applications.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the architecture and internal peripherals of the 8051 microcontroller.	Understand
CO2	Develop microcontroller programs for interfacing sensors, displays and actuators.	Apply
CO3	Explain PLC hardware, operation and industrial automation architectures.	Understand
CO4	Develop Ladder Logic programs for industrial sequence control and automation.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	8051 Microcontroller Architecture	Introduction to microcontrollers; 8051 architecture: CPU, registers, memory organization (RAM/ROM), Special Function Registers (SFRs); I/O ports; Oscillator and reset circuits.	Approx. 14 hours	CO1	Module 1
2	8051 Programming and Interfacing	Addressing modes; Instruction set (Data transfer, Arithmetic, Logical, Branching); Assembly and C programming; Interfacing: LED, LCD, 4x4 Keypad, ADC/DAC, and DC motor control.	Approx. 16 hours	CO2	Module 2
3	PLC Fundamentals and Hardware	Introduction to PLCs; Advantages over relay logic; PLC architecture: CPU, Power supply, I/O modules (Digital/Analog); Scan cycle: Read, Execute, Write; PLC communication networks.	Approx. 14 hours	CO3	Module 3
4	Ladder Logic Programming and Applications	Programming languages (IEC 61131-3); Ladder Logic (LD) fundamentals: Rungs, Rails, NO/NC contacts, Coils; Timers (TON, TOF) and Counters (CTU, CTD); Industrial applications: Motor starter, Conveyor control, Traffic light.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

8051 Microcontroller Architecture

Introduction to microcontrollers; 8051 architecture: CPU, registers, memory organization (RAM/ROM), Special Function Registers (SFRs); I/O ports; Oscillator and reset circuits.

CO1 · Approx. 14 hours

8051 Programming and Interfacing

Addressing modes; Instruction set (Data transfer, Arithmetic, Logical, Branching); Assembly and C programming; Interfacing: LED, LCD, 4x4 Keypad, ADC/DAC, and DC motor control.

CO2 · Approx. 16 hours

PLC Fundamentals and Hardware

Introduction to PLCs; Advantages over relay logic; PLC architecture: CPU, Power supply, I/O modules (Digital/Analog); Scan cycle: Read, Execute, Write; PLC communication networks.

CO3 · Approx. 14 hours

Ladder Logic Programming and Applications

Programming languages (IEC 61131-3); Ladder Logic (LD) fundamentals: Rungs, Rails, NO/NC contacts, Coils; Timers (TON, TOF) and Counters (CTU, CTD); Industrial applications: Motor starter, Conveyor control, Traffic light.

CO4 · Approx. 16 hours

MODULE 1

8051 Microcontroller Architecture

Introduction to microcontrollers; 8051 architecture: CPU, registers, memory organization (RAM/ROM), Special Function Registers (SFRs); I/O ports; Oscillator and reset circuits.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

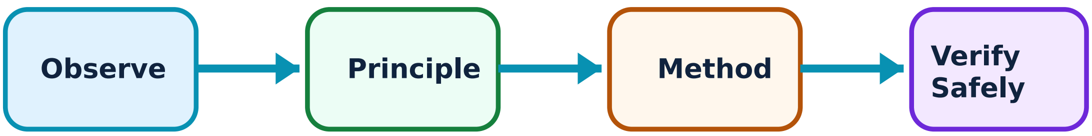
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 8051 Microcontroller Architecture: identify the system, state its principle, follow the correct method and verify the result safely.

1. 8051 internal structure−

The 8051 is a popular 8-bit microcontroller with an on-chip CPU, RAM, ROM, timers, and serial port. Its memory is divided into Program Memory (ROM) and Data Memory (RAM). SFRs are used to control internal peripherals like timers and I/O ports.

Study and application method

Sketch the block diagram of the 8051 microcontroller; explain the function of the 'Program Status Word' (PSW) register.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the block diagram of the 8051 microcontroller; explain the function of the 'Program Status Word' (PSW) register.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Microcontrollers are sensitive to static electricity and voltage fluctuations. Do not handle or power the device without authorised ESD protection and regulated power supplies.

2. Memory and I/O organization−

8051 uses a Harvard architecture with separate address spaces for code and data. It has four 8-bit I/O ports (P0-P3). Port 0 and Port 2 are often used for external memory interfacing, while Port 3 has alternate functions like interrupts and serial communication.

Study and application method

Describe the internal RAM organization of 8051; explain the 'Dual Function' of Port 3 pins.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the internal RAM organization of 8051; explain the 'Dual Function' of Port 3 pins.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper pin connections can damage the microcontroller. Do not connect high-current loads directly to I/O ports without authorised buffer circuits or transistors.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

8051 Programming and Interfacing

Addressing modes; Instruction set (Data transfer, Arithmetic, Logical, Branching); Assembly and C programming; Interfacing: LED, LCD, 4x4 Keypad, ADC/DAC, and DC motor control.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

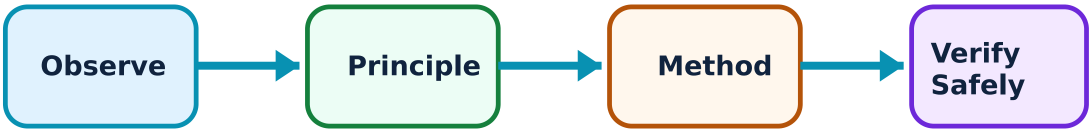
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 8051 Programming and Interfacing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Instruction set and programming−

8051 supports various addressing modes like immediate, register, and direct. Programming can be done in Assembly for low-level control or C for ease of development. Timers and interrupts are used for precise timing and event handling.

Study and application method

Write a simple 8051 C program to blink an LED connected to P1.0; explain the difference between 'Timer' and 'Counter' modes.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write a simple 8051 C program to blink an LED connected to P1.0; explain the difference between 'Timer' and 'Counter' modes.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Infinite loops or incorrect interrupt handling can cause system hangs. All control software must follow authorised coding standards and include watchdog timer resets where applicable.

2. Peripheral interfacing−

Interfacing involves connecting external devices to the microcontroller. LCDs provide visual feedback, keypads allow user input, and ADCs convert analog sensor signals to digital. PWM (Pulse Width Modulation) is used to control the speed of DC motors.

Study and application method

Sketch the interface circuit for a 16x2 LCD with 8051; explain the steps to read data from a 4x4 matrix keypad.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the interface circuit for a 16x2 LCD with 8051; explain the steps to read data from a 4x4 matrix keypad.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Interfacing high-voltage AC loads requires isolation. Do not interface AC devices without authorised opto-isolators, relays, and proper grounding.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

PLC Fundamentals and Hardware

Introduction to PLCs; Advantages over relay logic; PLC architecture: CPU, Power supply, I/O modules (Digital/Analog); Scan cycle: Read, Execute, Write; PLC communication networks.

Approx. 14 hours **CO3** **Understand · Apply**

Learning goals

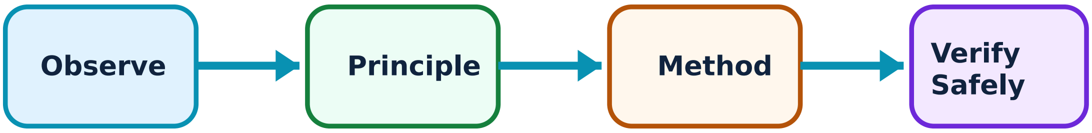
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PLC Fundamentals and Hardware: identify the system, state its principle, follow the correct method and verify the result safely.

1. PLC architecture and operation−

A PLC is a ruggedized computer used for industrial automation. It consists of a CPU, I/O modules, and a power supply. The 'Scan Cycle' is the repetitive process where the PLC reads inputs, executes the user program, and updates outputs.

Study and application method

Compare PLCs with traditional Relay Logic systems; explain the three main phases of a 'PLC Scan Cycle'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare PLCs with traditional Relay Logic systems; explain the three main phases of a 'PLC Scan Cycle'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: PLC systems operate in harsh industrial environments. Do not install or modify PLC hardware without authorised enclosures, surge protection, and adherence to industrial wiring codes.

2. I/O modules and wiring−

I/O modules bridge the PLC and the industrial process. Digital modules handle ON/OFF signals (switches, solenoids), while Analog modules handle continuous signals (temperature, pressure). Proper wiring and addressing are critical for system operation.

Study and application method

Differentiate between 'Sinking' and 'Sourcing' I/O configurations; describe the role of a 'Remote I/O' system in large factories.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Sinking' and 'Sourcing' I/O configurations; describe the role of a 'Remote I/O' system in large factories.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect I/O wiring can lead to machine malfunction or injury. All PLC I/O connections must follow authorised wiring diagrams and be verified before power-up.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Ladder Logic Programming and Applications

Programming languages (IEC 61131-3); Ladder Logic (LD) fundamentals: Rungs, Rails, NO/NC contacts, Coils; Timers (TON, TOF) and Counters (CTU, CTD); Industrial applications: Motor starter, Conveyor control, Traffic light.

Approx. 16 hours | **CO4** | **Understand · Apply**

Learning goals

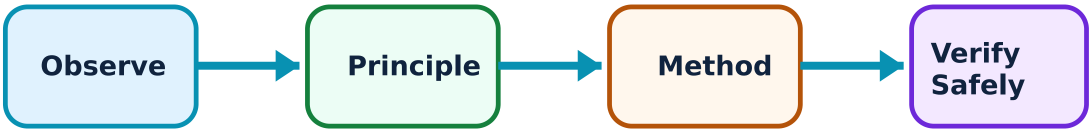
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Ladder Logic Programming and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. Ladder Logic fundamentals−

Ladder Logic is a graphical programming language that resembles electrical relay diagrams. Rungs represent logical paths. Contacts (NO/NC) represent inputs, and Coils represent outputs. Timers and Counters add time-based and event-based control.

Study and application method

Draw the Ladder Logic for a 'Start-Stop' motor circuit with latching; explain the operation of an 'On-Delay Timer' (TON).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the Ladder Logic for a 'Start-Stop' motor circuit with latching; explain the operation of an 'On-Delay Timer' (TON).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Logic errors in Ladder programs can cause physical damage. Do not run new PLC programs on live machinery without authorised simulation and dry-run testing.

2. Industrial automation applications−

PLCs are used to automate complex sequences. Examples include Star-Delta starters for motors, conveyor belt sorting systems, and automated traffic control. Integration with SCADA (Supervisory Control and Data Acquisition) allows for centralized monitoring.

Study and application method

Sketch a Ladder Diagram for a three-aspect traffic light controller; describe how a PLC interfaces with a SCADA system conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a Ladder Diagram for a three-aspect traffic light controller; describe how a PLC interfaces with a SCADA system conceptually.

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Common mistake / precaution: Automated systems must include physical emergency stops (E-Stops). Do not rely solely on PLC software for safety-critical shutdowns; always use authorised hard-wired safety circuits.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: 8051 Microcontroller Architecture

Use the concept flow to revise observation, principle, method and verification.

Module 2: 8051 Programming and Interfacing

Use the concept flow to revise observation, principle, method and verification.

Module 3: PLC Fundamentals and Hardware

Use the concept flow to revise observation, principle, method and verification.

Module 4: Ladder Logic Programming and Applications

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Microcontrollers and PLCs in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Write an 8051 C code snippet to generate a 1ms delay using Timer 0.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a Ladder Diagram for a simple 'OR' logic gate using two switches and one lamp.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the PLC I/O wiring for a motor controlled by a push button and a limit switch.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a PLC-based automation system in a local industry or a video demonstration (e.g., bottling plant).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — 8051 Microcontroller Architecture

Explain 8051 internal structure and state one correct method, application or precaution.−

The 8051 is a popular 8-bit microcontroller with an on-chip CPU, RAM, ROM, timers, and serial port. Its memory is divided into Program Memory (ROM) and Data Memory (RAM). SFRs are used to control internal peripherals like timers and I/O ports.

Answer framework: Sketch the block diagram of the 8051 microcontroller; explain the function of the 'Program Status Word' (PSW) register.

Important point: Microcontrollers are sensitive to static electricity and voltage fluctuations. Do not handle or power the device without authorised ESD protection and regulated power supplies.

Explain Memory and I/O organization and state one correct method, application or precaution.−

8051 uses a Harvard architecture with separate address spaces for code and data. It has four 8-bit I/O ports (P0-P3). Port 0 and Port 2 are often used for external memory interfacing, while Port 3 has alternate functions like interrupts and serial communication.

Answer framework: Describe the internal RAM organization of 8051; explain the 'Dual Function' of Port 3 pins.

Important point: Improper pin connections can damage the microcontroller. Do not connect high-current loads directly to I/O ports without authorised buffer circuits or transistors.

Module 2 — 8051 Programming and Interfacing

Explain Instruction set and programming and state one correct method, application or precaution.−

8051 supports various addressing modes like immediate, register, and direct. Programming can be done in Assembly for low-level control or C for ease of development. Timers and interrupts are used for precise timing and event handling.

Answer framework: Write a simple 8051 C program to blink an LED connected to P1.0; explain the difference between 'Timer' and 'Counter' modes.

Important point: Infinite loops or incorrect interrupt handling can cause system hangs. All control software must follow authorised coding standards and include watchdog timer resets where applicable.

Explain Peripheral interfacing and state one correct method, application or precaution.−

Interfacing involves connecting external devices to the microcontroller. LCDs provide visual feedback, keypads allow user input, and ADCs convert analog sensor signals to digital. PWM (Pulse Width Modulation) is used to control the speed of DC motors.

Answer framework: Sketch the interface circuit for a 16x2 LCD with 8051; explain the steps to read data from a 4x4 matrix keypad.

Important point: Interfacing high-voltage AC loads requires isolation. Do not interface AC devices without authorised opto-isolators, relays, and proper grounding.

Module 3 — PLC Fundamentals and Hardware

Explain PLC architecture and operation and state one correct method, application or precaution.−

A PLC is a ruggedized computer used for industrial automation. It consists of a CPU, I/O modules, and a power supply. The 'Scan Cycle' is the repetitive process where the PLC reads inputs, executes the user program, and updates outputs.

Answer framework: Compare PLCs with traditional Relay Logic systems; explain the three main phases of a 'PLC Scan Cycle'.

Important point: PLC systems operate in harsh industrial environments. Do not install or modify PLC hardware without authorised enclosures, surge protection, and adherence to industrial wiring codes.

Explain I/O modules and wiring and state one correct method, application or precaution.−

I/O modules bridge the PLC and the industrial process. Digital modules handle ON/OFF signals (switches, solenoids), while Analog modules handle continuous signals (temperature, pressure). Proper wiring and addressing are critical for system operation.

Answer framework: Differentiate between 'Sinking' and 'Sourcing' I/O configurations; describe the role of a 'Remote I/O' system in large factories.

Important point: Incorrect I/O wiring can lead to machine malfunction or injury. All PLC I/O connections must follow authorised wiring diagrams and be verified before power-up.

Module 4 — Ladder Logic Programming and Applications

Explain Ladder Logic fundamentals and state one correct method, application or precaution.−

Ladder Logic is a graphical programming language that resembles electrical relay diagrams. Rungs represent logical paths. Contacts (NO/NC) represent inputs, and Coils represent outputs. Timers and Counters add time-based and event-based control.

Answer framework: Draw the Ladder Logic for a 'Start-Stop' motor circuit with latching; explain the operation of an 'On-Delay Timer' (TON).

Important point: Logic errors in Ladder programs can cause physical damage. Do not run new PLC programs on live machinery without authorised simulation and dry-run testing.

Explain Industrial automation applications and state one correct method, application or precaution.−

PLCs are used to automate complex sequences. Examples include Star-Delta starters for motors, conveyor belt sorting systems, and automated traffic control. Integration with SCADA (Supervisory Control and Data Acquisition) allows for centralized monitoring.

Answer framework: Sketch a Ladder Diagram for a three-aspect traffic light controller; describe how a PLC interfaces with a SCADA system conceptually.

Important point: Automated systems must include physical emergency stops (E-Stops). Do not rely solely on PLC software for safety-critical shutdowns; always use authorised hard-wired safety circuits.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: 8051 Microcontroller Architecture.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: 8051 Programming and Interfacing.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: PLC Fundamentals and Hardware.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Ladder Logic Programming and Applications.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Introduction to microcontrollers; 8051 architecture: CPU, registers, memory organization (RAM/ROM), Special Function Registers (SFRs); I/O ports; Oscillator and reset circuits..
2. Develop a complete answer or practical plan for: Addressing modes; Instruction set (Data transfer, Arithmetic, Logical, Branching); Assembly and C programming; Interfacing: LED, LCD, 4x4 Keypad, ADC/DAC, and DC motor control..
3. Develop a complete answer or practical plan for: Introduction to PLCs; Advantages over relay logic; PLC architecture: CPU, Power supply, I/O modules (Digital/Analog); Scan cycle: Read, Execute, Write; PLC communication networks..
4. Develop a complete answer or practical plan for: Programming languages (IEC 61131-3); Ladder Logic (LD) fundamentals: Rungs, Rails, NO/NC contacts, Coils; Timers (TON, TOF) and Counters (CTU, CTD); Industrial applications: Motor starter, Conveyor control, Traffic light..

LAST-MILE REVISION

Quick Revision

Module 1: 8051 Microcontroller Architecture

Introduction to microcontrollers; 8051 architecture: CPU, registers, memory organization (RAM/ROM), Special Function Registers (SFRs); I/O ports; Oscillator and reset circuits.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: 8051 Programming and Interfacing

Addressing modes; Instruction set (Data transfer, Arithmetic, Logical, Branching); Assembly and C programming; Interfacing: LED, LCD, 4x4 Keypad, ADC/DAC, and DC motor control.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: PLC Fundamentals and Hardware

Introduction to PLCs; Advantages over relay logic; PLC architecture: CPU, Power supply, I/O modules (Digital/Analog); Scan cycle: Read, Execute, Write; PLC communication networks.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Ladder Logic Programming and Applications

Programming languages (IEC 61131-3); Ladder Logic (LD) fundamentals: Rungs, Rails, NO/NC contacts, Coils; Timers (TON, TOF) and Counters (CTU, CTD); Industrial applications: Motor starter, Conveyor control, Traffic light.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
8051 internal structure	The 8051 is a popular 8-bit microcontroller with an on-chip CPU, RAM, ROM, timers, and serial port.
Memory and I/O organization	8051 uses a Harvard architecture with separate address spaces for code and data.
Instruction set and programming	8051 supports various addressing modes like immediate, register, and direct.
Peripheral interfacing	Interfacing involves connecting external devices to the microcontroller.
PLC architecture and operation	A PLC is a ruggedized computer used for industrial automation.
I/O modules and wiring	I/O modules bridge the PLC and the industrial process.
Ladder Logic fundamentals	Ladder Logic is a graphical programming language that resembles electrical relay diagrams.
Industrial automation applications	PLCs are used to automate complex sequences.

LEARNING RESOURCES

References

Recommended microcontroller and PLC references

1. Muhammad Ali Mazidi, The 8051 Microcontroller and Embedded Systems.
2. Kenneth J. Ayala, The 8051 Microcontroller.
3. Frank D. Petruzella, Programmable Logic Controllers.
4. John W. Webb and Ronald A. Reis, Programmable Logic Controllers: Principles and Applications.

Approved public microcontroller and PLC sources

- https://nptel.ac.in/courses/108105062
- https://onlinecourses.nptel.ac.in/noc21_me67/preview

These sources provide the verified study basis while official Course 6031T detail is unavailable; they do not replace official industrial designs, authorised PLC safety protocols, certified equipment specifications or authorised factory plans.